

PUBLICATION MATERIAL

Publication Plan and Manuscript Outline

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Publication Plan

Candidate theme

Selective CMOS Restoration in an FS-GDI Extended Hamming (12,7) Encoder: A Reproducible 65-nm Predictive LTspice Study

Evidence to use

- Independent reproduction of the Hamming (11,7) parity network.
- Exhaustive SEC-DED functional verification.
- CMOS, FS-GDI, Hybrid-A, and Hybrid-B transistor-level netlists.
- Nominal D1-to-P0 power, delay, PDP, energy, swing, and transistor counts.
- One-factor robustness sweeps and transparent rejected points.

Publication claim boundary

The paper may describe Hybrid-B as a speed-oriented compromise only. It must also report that FS-GDI has lower simulated power and PDP. Direct comparison with papers using a foundry PDK, different VDD, different loading, or post-layout extraction must be qualified.

Remaining work before submission

- Supervisor review of the reconstructed six-transistor FS-GDI topology.
- Optional independent HSPICE/Virtuoso cross-check.
- A broader systematic search in subscription databases.
- Language and similarity review under the university's rules.

Publication Manuscript Outline

- Introduction and verified research gap.
- Audited baseline and extended-code equations.
- CMOS, FS-GDI, Hybrid-A, and selective-output Hybrid-B circuits.
- PTM 65-nm LTspice methodology and traceability.
- XOR-level results.
- Hamming (11,7) reproduction.
- Extended Hamming nominal and robustness results.
- Speed-energy-restoration trade-off.
- Limitations and conclusion.

Provisional central result: Hybrid-B is the fastest tested extended encoder at the nominal condition and improves all reported metrics relative to CMOS, but unbuffered FS-GDI remains more energy efficient. The final title should emphasize selective restoration and trade-off rather than generic superiority.